

TASK 15 — CAPSTONE PROJECT (OPEN SOURCE INTELLIGENCE - OSINT)

Project Title: Comprehensive Cyber Reconnaissance and Threat Intelligence Audit on Mozilla Corporation

Target Organization: Mozilla Corporation (`mozilla.org`)

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GitHub Repository: <https://bit.ly/44U57sF>

Final Internship Project Report

This report is submitted as the final capstone project to fulfill the requirements of the OSINT Internship Program at DevShieldX.

1. Executive Summary

This report presents a detailed cybersecurity reconnaissance and Open Source Intelligence (OSINT) investigation on Mozilla Corporation (mozilla.org). Using only passive and safe active footprinting techniques, we mapped the target's internet-facing assets, servers, open ports, employee email patterns, and historical data. The investigation shows that Mozilla maintains a very strong security posture. Additionally, network infrastructure analysis revealed that Mozilla leverages Fastly's Content Delivery Network (CDN) and Anycast routing infrastructure to optimize global traffic delivery, ensure high availability, and mitigate volumetric Distributed Denial of Service (DDoS) attacks. However, several data fragments, exposed subdomains, and indexed documents are visible online, which could help an attacker map their internal network structure.

2. Objective

The main goals of this practical reconnaissance task are:

- * To map out publicly accessible servers, subdomains, and technical endpoints of the target organization.
- * To practice using industry-standard OSINT tools and search engines to gather threat intelligence.
- * To discover corporate communication channels, employee email formats, and operational assets.
- * To analyze the gathered data and write defensive security recommendations from a professional analyst's point of view.

Scope

The scope of this project is strictly limited to information gathering on Mozilla's primary domain (`mozilla.org`), its active subdomains, publicly exposed network ports, public code repositories (GitHub), and historical website archives. No aggressive scanning, exploit launches, or credential-guessing attempts were performed during this research.

4. Investigation Methodology

We followed a clean, step-by-step information-gathering framework:

- **Passive Infrastructure Discovery:** Checked public WHOIS domain registries and analyzed DNS routing patterns to find active servers.
- **Cyber Asset Search Engines:** Used Shodan and VirusTotal to check the reputation of the domain and find open network ports.
- **Subdomain & Web Application Scraping:** Gathered and sorted subdomains to find different active projects run by the company.
- **Human-Centric Social OSINT:** Analyzed corporate public profiles and email listing structures to understand the organization's workforce setup.
- **Historical Data Analysis:** Used the Wayback Machine to find older files, documentation, and historical infrastructure changes.

5. Target Profile

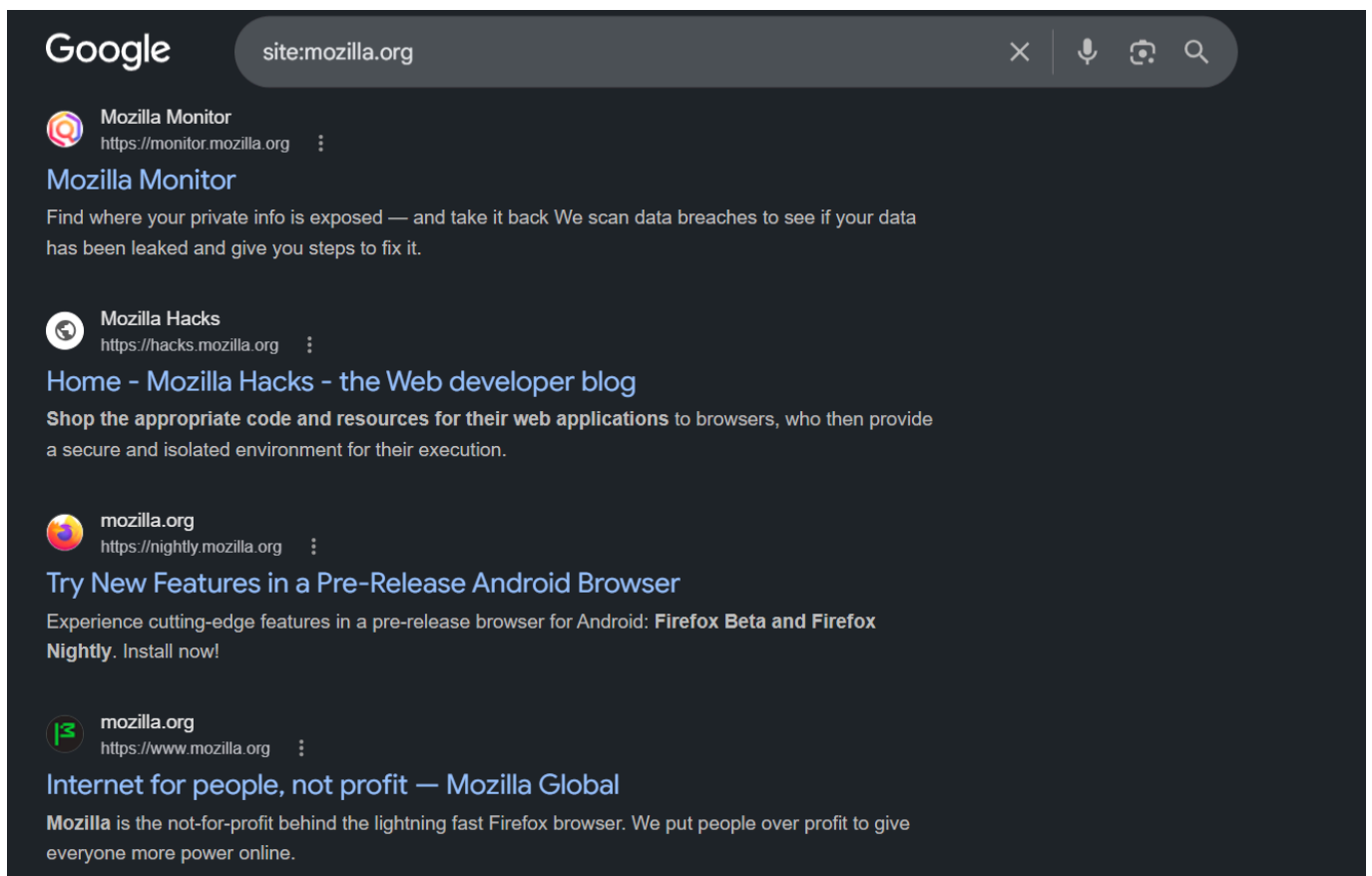
- **Company Name:** Mozilla Corporation / Mozilla Foundation
- **Main Domain Name:** `mozilla.org`
- **Industry Type:** Software Engineering, Web Browsers, and Open-Source Internet Tools
- **Corporate Location:** San Francisco, California, United States
- **Primary Products:** Firefox Browser, Mozilla Monitor, MDN Web Docs, Bugzilla, Pocket

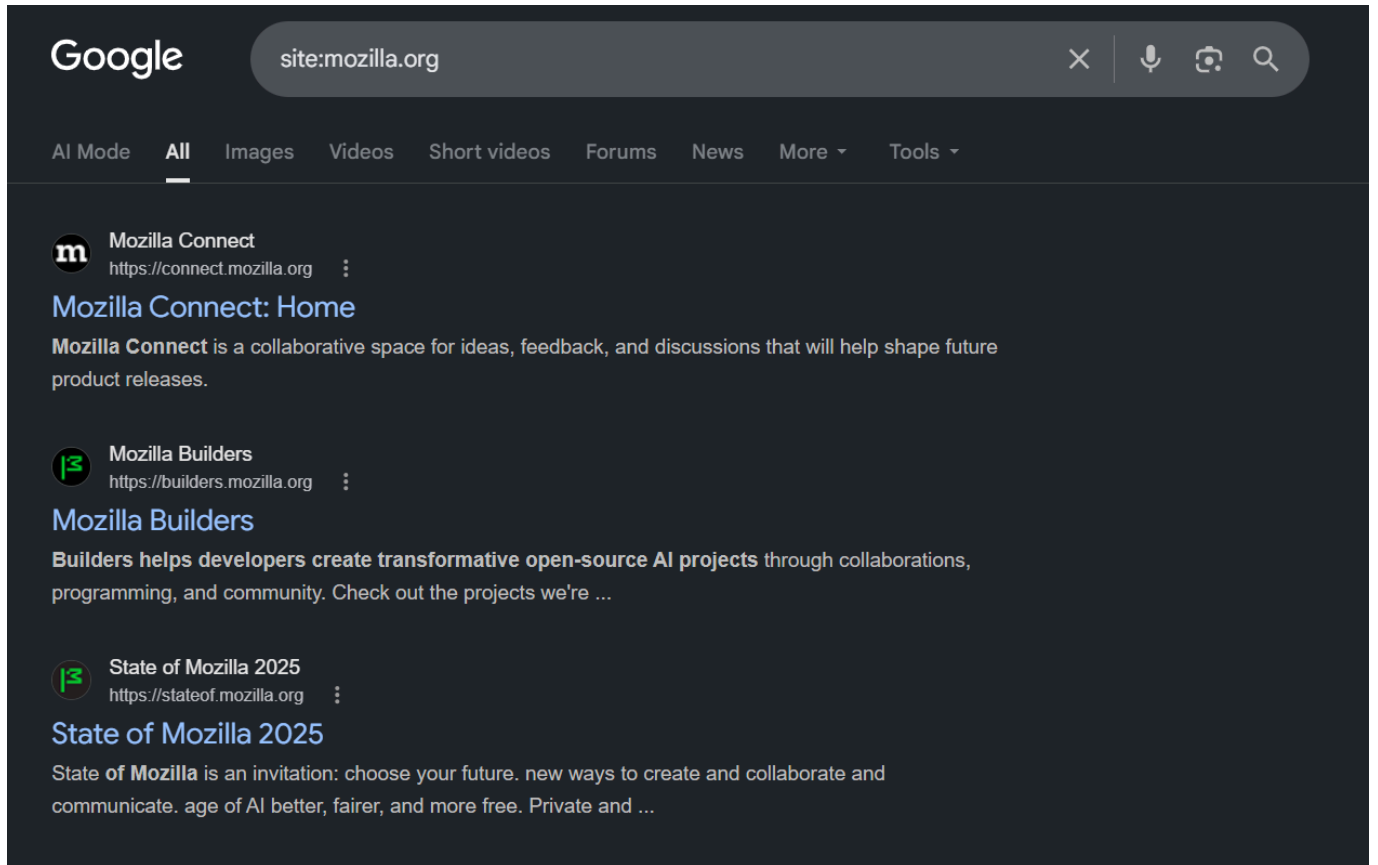
6. Digital Footprint Analysis

6.1. Subdomain Mapping

Mozilla manages a massive online network with many micro-services. By studying active subdomain feeds, we identified multiple operational systems:

- **Telemetry Systems:** Domains like `incoming.telemetry.mozilla.org` and `missioncontrol.telemetry.mozilla.org` collect browser performance metrics and crash logs.
- **Internal Authentication Pointers:** Endpoints such as `ldap.mozilla.org` and `passwordreset.mozilla.org` handle internal staff directory entries and authentication.





6.2. Corporate Email Harvesting & Format Logic

Understanding employee email layouts is highly valuable during the initial phases of a security audit. Our corporate search revealed two standard setups used by the company:

Domain Standard Pattern (LeadIQ Findings): Mozilla uses two active mail servers:

For `mozilla.com`: The dominant format is `FLast@mozilla.com` (First Initial + Last Name, making up 89% of accounts).

For `mozilla.org`: The dominant format is `First@mozilla.org` (First Name only, making up 51% of accounts).

Live Target Verification: Automated validation engines successfully confirmed several real-time live emails belonging to active workers (e.g., `sarah@mozilla.org`, `selgamal@mozilla.org`, and `yamini@mozilla.org`).

Mozilla Email Formats

Example

Percentage

FLast@mozilla.com	JDoe@mozilla.com	89%
FirstLast@mozilla.com	JohnDoe@mozilla.com	6%
First@mozilla.com	John@mozilla.com	3%
FL@mozilla.com	JD@mozilla.com	2%
First@mozilla.org	John@mozilla.org	51%
First.Last@mozilla.org	John.Doe@mozilla.org	28%
FirstLast@mozilla.org	JohnDoe@mozilla.org	13%
Last@mozilla.org	Doe@mozilla.org	8%



Find emails:

by Name

by Domain

from LinkedIn

Company (domain or name)

mozilla.org

Find email →

sarah@mozilla.org

Verified ✓

selgamal@mozilla.org

Verified ✓

yamini@mozilla.org

Verified ✓

max@mozilla.org

Verified ✓

alicia@mozilla.org

Verified ✓

eric@mozilla.org	Verified ✓
mark@mozilla.org	Verified ✓
morgan@mozilla.org	Verified ✓
alia@mozilla.org	Verified ✓
charlie@mozilla.org	Verified ✓
camilla@mozilla.org	Verified ✓
amy@mozilla.org	Verified ✓
asa@mozilla.org	Verified ✓

7. Infrastructure Intelligence

7.1. Global Server Spread

According to current Shodan asset intelligence database entries, Mozilla has ****848 distinct assets**** connected directly to the internet.

Geographic Distribution: The United States hosts the majority of their systems with 343 active machines. This is followed by the United Kingdom (70), Germany (43), Japan (38), and China (36).

7.2. Hosting Providers & Content Delivery Networks (CDNs)

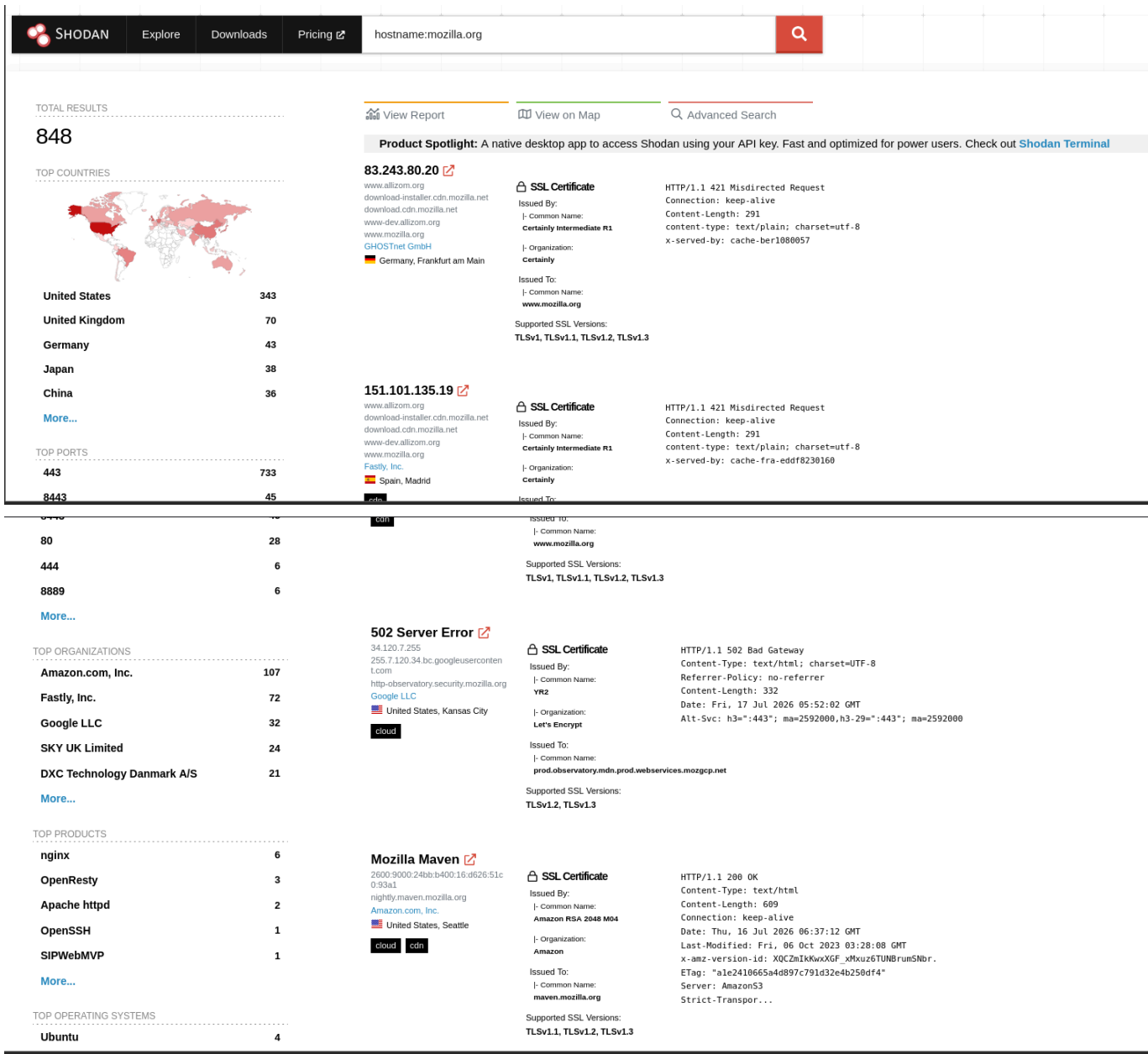
To maintain fast global loading times and block volumetric DDoS attacks, Mozilla distributes its platform load across major tech providers:

- **Amazon.com, Inc. (107 hosts):** Manages primary cloud application storage blocks and web nodes.
- **Fastly, Inc. (72 hosts):** Handles global caching layers to route end-user web traffic quickly.
- **Google LLC (32 hosts):** Manages processing systems and secondary cloud hosting environments.

7.3. Active Network Port Exposure

The network scan reveals which ports are actively listening on Mozilla's public assets:

- **Port 443 (HTTPS):** 733 open hosts. This indicates that almost all their web servers enforce modern transport layer encryption (TLS).
- **Port 8443 (Alternative HTTPS):** 45 hosts. This port is generally utilized for alternative administrative web tools or testing dashboards.
- **Port 80 (HTTP):** 28 hosts. These handle basic legacy web requests and automatically redirect users to secure channels.



8. Technology Stack Analysis

A deeper look into their public infrastructure reveals the software frameworks running behind the scenes:

- **Web Proxy & Servers:** Mozilla relies heavily on `nginx`, `OpenResty`, and `Apache httpd` to route inbound traffic. Remote administrative shell connectivity runs through standard secure systems (`OpenSSH`).
- **Cloud Storage Containers:** Public release systems and code binaries are hosted on Amazon S3 storage infrastructure (`nightly.maven.mozilla.org`), which returns safe web response codes (HTTP 200 OK).
- **Encryption Standards:** Their web applications protect user communication by using verified certificates issued by **Let's Encrypt** and **Amazon RSA**. These certificates actively support TLS version 1.2 and version 1.3 cryptographic protocols.

80	28	cloud	Issued To: - Common Name: www.mozilla.org	
444	6		Supported SSL Versions: TLSv1, TLSv1.1, TLSv1.2, TLSv1.3	
8889	6			
More...				
TOP ORGANIZATIONS				
Amazon.com, Inc.	107		34.120.7.255 255.7.120.34.bc.googleusercontent.com http-observatory.security.mozilla.org	SSL Certificate Issued By: - Common Name: YNZ - Organization: Let's Encrypt Issued To: - Common Name: prod.observatory.mdn.prod.webservices.mozgcp.net Supported SSL Versions: TLSv1.2, TLSv1.3
Fastly, Inc.	72		Google LLC United States, Kansas City	HTTP/1.1 502 Bad Gateway Content-Type: text/html; charset=UTF-8 Referrer-Policy: no-referrer Content-Length: 332 Date: Fri, 17 Jul 2026 05:52:02 GMT Alt-Svc: h3=":443"; ma=2592000, h3-29=":443"; ma=2592000
Google LLC	32	cloud		
SKY UK Limited	24			
DXC Technology Danmark A/S	21			
More...				
TOP PRODUCTS				
nginx	6		Mozilla Maven 2600:9000:24bb:b400:16:0626:51c0:93a1 nightly.maven.mozilla.org	SSL Certificate Issued By: - Common Name: Amazon RSA 2048 M04 - Organization: Amazon Issued To: - Common Name: maven.mozilla.org Supported SSL Versions: TLSv1.1, TLSv1.2, TLSv1.3
OpenResty	3		Amazon.com, Inc. United States, Seattle	HTTP/1.1 200 OK Content-Type: text/html Content-Length: 609 Connection: keep-alive Date: Thu, 16 Jul 2026 06:37:12 GMT Last-Modified: Fri, 06 Oct 2023 03:28:08 GMT x-amz-version-id: XQCZmIKKwXGF_xMxuz6TUNBumSNbr. ETag: "a1e2410665a4d897c791d32e4b250df4" Server: AmazonS3 Strict-Transpor...
Apache httpd	2			
OpenSSH	1			
SIPWebMVP	1			
More...				
TOP OPERATING SYSTEMS				
Ubuntu	4			

9. Social Media & Public Presence

9.1. Corporate Size Analysis

Checking public business profiles helped us cross-examine their overall organizational size:

Workforce Mapping: LinkedIn metrics report the corporate size category as 501-1,000 employees. However, there are approximately 1,772 registered users listed on the platform who are current employees at Mozilla.

9.2. Code Development Repositories (GitHub Audit)

Mozilla's public-facing engineering presence (`github.com/mozilla`)(<https://github.com/mozilla>) is incredibly massive and transparent:

They host extensive source code systems. Their primary development technologies include JavaScript, Rust, Python, HTML, and CSS.

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Mozilla

View all 1,772 employees

Mozilla

Software Development

San Francisco, CA · 447,724 followers

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Overview

Jobs

Life

About us

A lot of companies say they're "mission-driven." Our unique corporate structure guarantees that every decision we make upholds our mission: to ensure the internet remains open and accessible. Beholden to neither shareholders nor investors, Mozilla Corporation is wholly owned by the not-for-profit Mozilla Foundation. Along with our communities of 20,000+ contributors and collaborators, Mozilla Corporation's staff designs, builds, and distributes software that allows people to enjoy the internet on their own terms. Our flagship product, the Firefox browser, has expanded into a family of products

Pocket

Technology, Information and Internet

San Francisco, California

Mozilla Festival

Civic and Social Organizations

Firefox

Technology, Information and Internet

ODIN.ai

Computer and Network Security

Austin, Texas

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Similar pages

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Paoli, PA

Brave

Technology, Information and Internet

San Francisco, California

Proton

Technology, Information and Internet

Geneva, Geneva

GitLab

that protects users and alerts them of risks, safeguards passwords and provides a secure VPN (with more to come). By maintaining a safe, open internet we're helping humanity, while also helping the individual humans employed here to reach their personal and professional goals. With a relatively small team serving hundreds of millions of people, a culture of exploration, and a commitment to mentorship, opportunities abound to learn and grow at Mozilla.

Website

http://www.mozilla.org

Industry

Software Development

Company size

501-1,000 employees

Headquarters

San Francisco, CA

Type

Privately Held

Founded

2005

Specialties

browser, internet, software, mobile, web apps, OS, identity, android, data science, and open source

Browse jobs

Mozilla jobs

3,037 open jobs

Engineer jobs

555,845 open jobs

Director jobs

1,220,357 open jobs

Marketing Manager jobs

106,879 open jobs

Director of Communications jobs

19,981 open jobs

Manager jobs

1,880,925 open jobs

Developer jobs

10. Risk Assessment

Target Exposure Point	Evaluated Risk	Core Security Finding & Potential Impact
Public Email Harvesting	Medium	Valid email lists and internal corporate naming formats are completely visible online. This exposure facilitates highly targeted Spear-Phishing, social engineering, and Business Email Compromise (BEC) campaigns against corporate personnel.
Open Admin Ports (8443)	Low to Medium	Having 45 active hosts with alternative web management ports (8443) results in the potential exposure of alternative web management interfaces or

		development environments, expanding the external network attack surface. If these panels are unpatched or misconfigured, they could become critical targets for automated brute-force attacks.
Publicly Indexed PDFs	Low	Search engines have indexed multi-page development and architectural files. This documentation gives an attacker easy access to structural and application specifications.

11. Intelligence Findings

Advanced Google Dorking Exploits: Initial Google Dorking queries utilizing the inurl:login operator return indexed URLs primarily associated with public documentation, MDN Web Docs glossaries (e.g., discussions on SQL Injection and Threat Modeling), and community support forums. While no administrative credentials or sensitive internal login portals were bypassed or exposed, the visibility of these troubleshooting threads highlights the extensive footprint indexed by search engine web crawlers.

Threat Intel Reputation Health: Running the main domain through VirusTotal showed a completely clean status. Out of **92 top security scanners** (like Google Safe Browsing, Symantec, Kaspersky, and Sophos), zero vendors flagged the domain as malicious. The reputation score is entirely clean.

Google site:mozilla.org filetype:pdf

mozilla.org
https://wirelesschallenge.mozilla.org › _assets › N... PDF

Smart Community Networks Challenge
15-Jun-2017 — The Smart Community Networks Challenge aims to close this access gap by **enhancing wireless connectivity** in areas where Internet access is a ...

mozilla.org
https://wirelesschallenge.mozilla.org › _assets › N... PDF

Off-the-Grid Internet Challenge
15-Jun-2017 — The **Off-the-Grid Internet Challenge** seeks ideas that can provide connectivity. Off-the-Grid Internet Challenge seeks solutions that help ...

mozilla.org
https://hg.mozilla.org › file › brotli › enc › prefix PDF

Knappcells batterier (översättningstabel)
Knappcells batterier (översättningstabel) Cell Ucar/Energizer ... 11,6 × 5,4 SR44 V357 357

The Mozilla Blog
https://blog.mozilla.org › files › 2017/01 › Mozil... PDF

Survey X
EUSurvey is an online survey-management system built for the creation and publishing of globally accessible forms, such as user satisfaction surveys



The Mozilla Blog

<https://blog.mozilla.org> › files › Questionnaire_mozilla

EUSurvey - The Mozilla Blog

25-Jun-2018 — **EUSurvey** is an online survey-management system built for the creation and publishing of globally accessible forms,



Mozilla

<https://www-archive.mozilla.org> › jsapi PDF

Chapter 1 Overview of the JavaScript C Engine ... - Mozilla

Chapter 1 Overview of the JavaScript C Engine1. How Do You Use the Engine?

How Does the Engine Relate to Applications? Managing a Run Time

178 pages



mozilla.org

<https://addons.mozilla.org> › firefox › user PDF

Reservaciones Delta Airlines: ¿Hablar con ...

25-May-2026 — +1-866-591-8384 | ¿Cómo hablar con alguien en Delta? Para hablar con alguien en Delta, debes comunicarte directamente con el servicio de ... [Read more](#)



Mozilla

<https://www-archive.mozilla.org> › security › iSEC... PDF

MOZILLA PHISHING PROTECTION

09-Nov-2006 — **Phishing Protection** feature, which will attempt to warn users of fraudulent websites impersonating legitimate ones. Mozilla added a new ...



Mozilla

<https://dictionary.telemetry.mozilla.org> › app_ids › tables

Table metrics for org.mozilla.firefox - Glean Dictionary

It is intended to be used together with `read\_query\_error\_count` to measure the overall error rate of read operations on the logins store.



Firefox Add-ons

<https://addons.mozilla.org> › en-US › firefox › addon › o...

ORganizer for Salesforce - Firefox for Android extensions

09-Oct-2024 — The ORganizer for Salesforce Extension lets you forget about your **Salesforce.com® username and passwords** and help you to recognize Salesforce. [Read more](#)

4.7 ★★★★★ (18) · Free



Bugzilla@Mozilla

<https://bugzilla.mozilla.org> › show_bug

111117 - Username and password appear in "Index of..." language

This component deals with FTP specific issues like downloading, directory formats, problems with FTP login, or resuming partial downloads, etc. Examples of ...

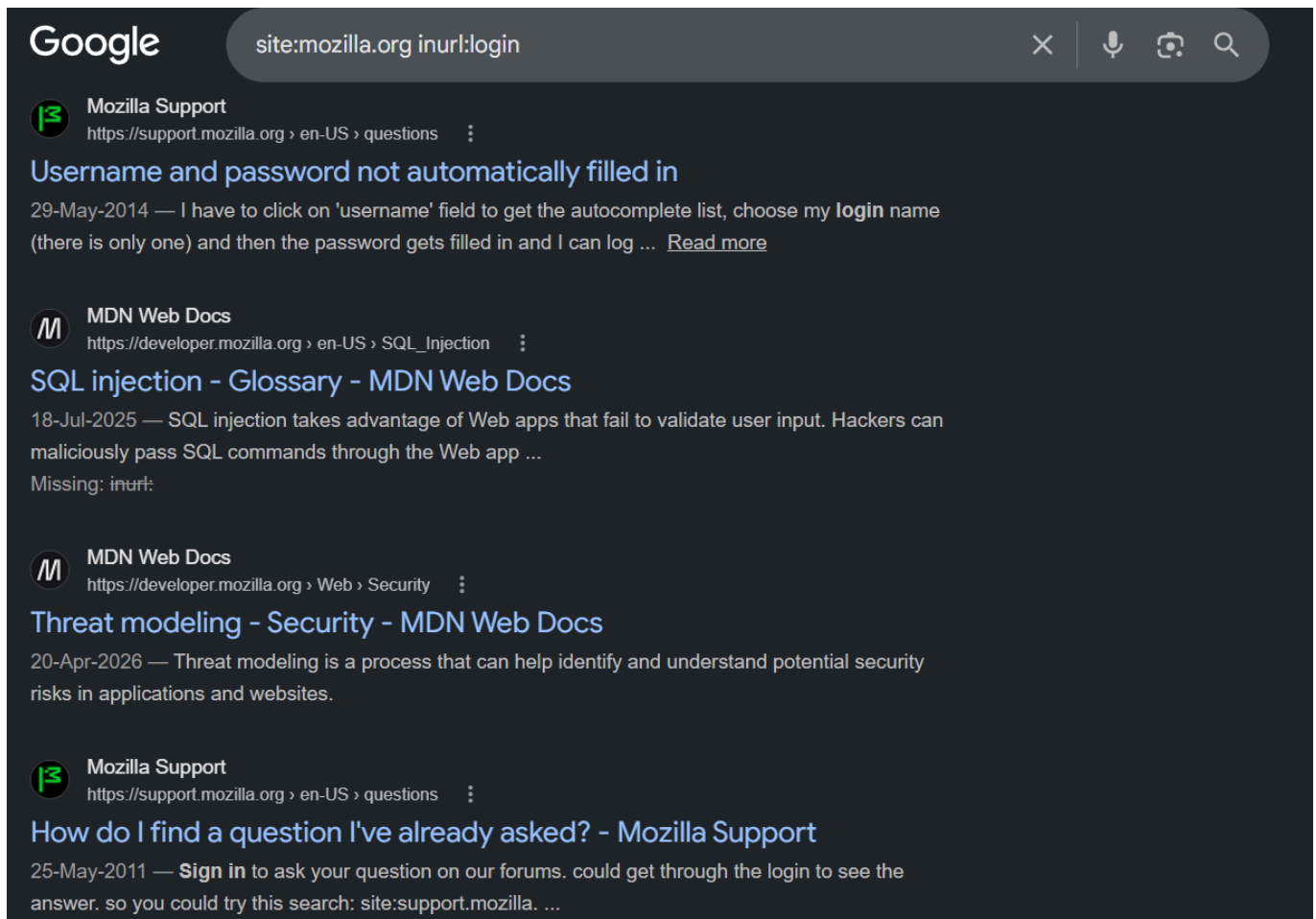


MDN Web Docs

<https://developer.mozilla.org> › ... › Client-side web APIs

Client-side storage - Learn web development | MDN

30-Oct-2025 — It consists of JavaScript APIs that allow you to store data on the client (ie on the user's machine) and then retrieve it when needed. [Read more](#)



12. Recommendations

12.1. Optimize Search Engine Indexing Rules:

Ensure that sensitive development paths, staging areas, and forums are restricted from search engine crawlers by configuring proper directives in the `robots.txt` file. Additionally, deploy the `X-Robots-Tag: noindex` HTTP header across non-public web pages to enforce strict anti-indexing rules globally and prevent the exposure of internal questionnaires or development logs.

12.2. Obfuscate Public Corporate Contact Details:

Enforce corporate email address obfuscation on public-facing sites or replace plain-text contact emails with secure contact forms protected by advanced CAPTCHA systems. Implement strict rate-limiting on directory endpoints to prevent automated scraping tools from harvesting employee communication data and initiating targeted phishing campaigns.

12.3. Secure Alternative Web Connections:

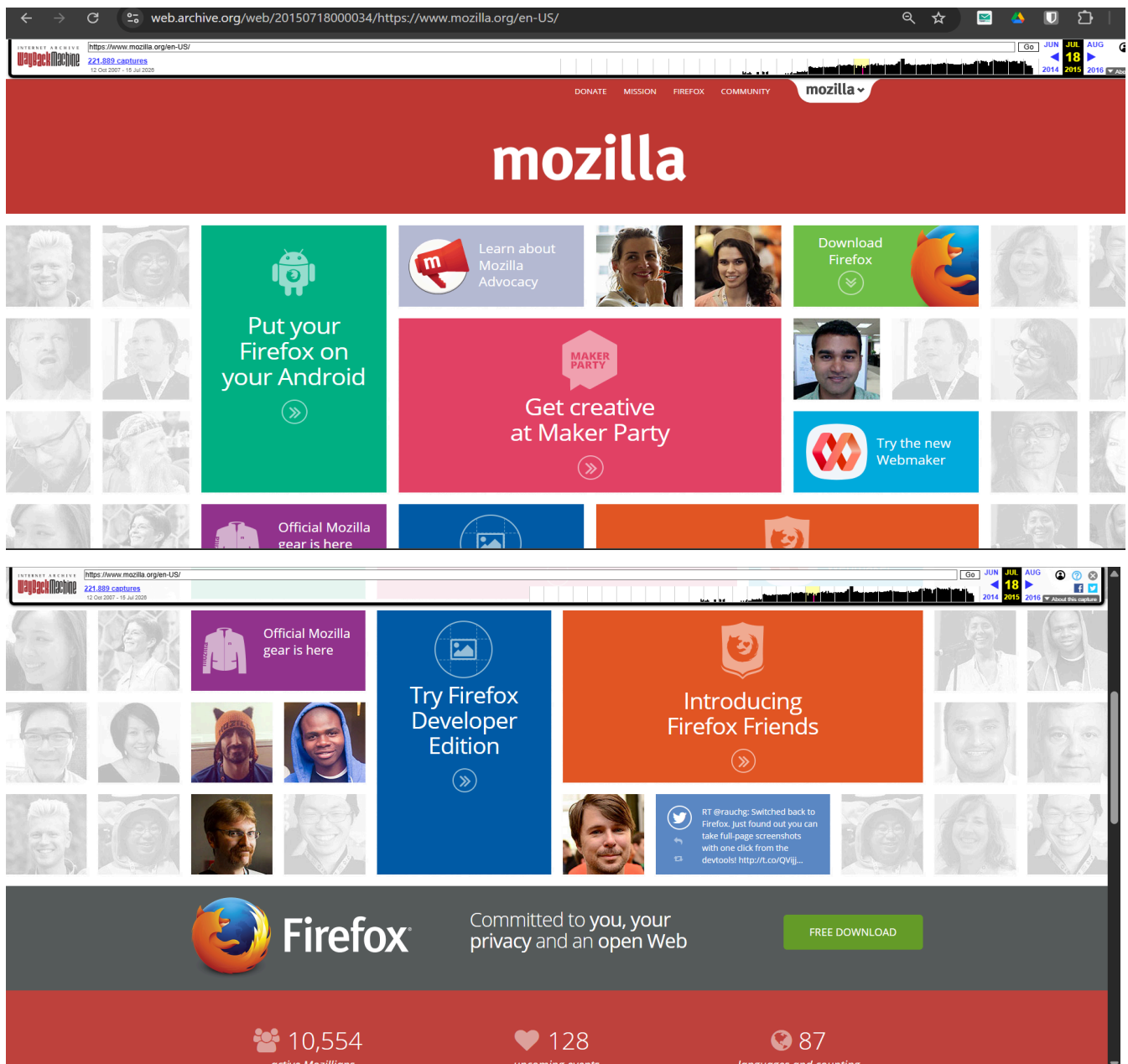
Restricting access to active dashboards on alternative ports, such as Port 8443, is critical. These management interfaces must be restricted behind an enterprise-grade Virtual Private Network (VPN) or Zero Trust Network Access (ZTNA) gateway, and protected by mandatory Multi-Factor Authentication (MFA) to mitigate unauthorized access and brute-force attempts.

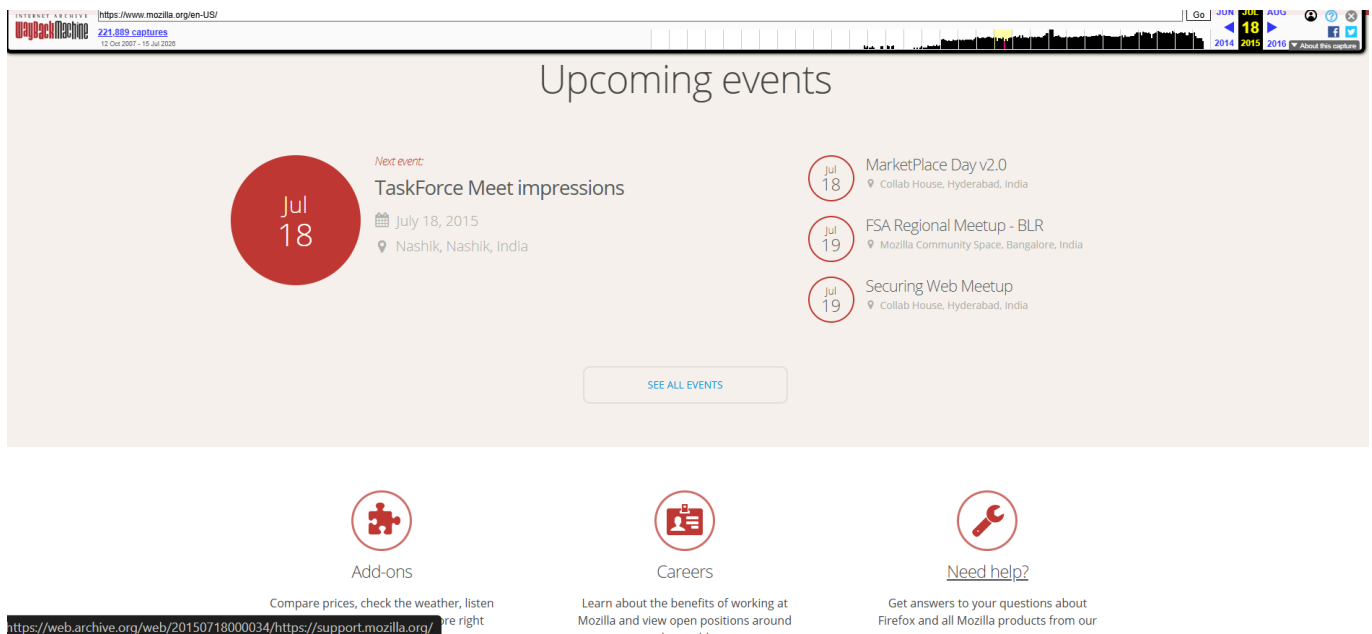
13. Conclusion

In conclusion, Mozilla Corporation (`mozilla.org`) maintains an exceptional, well-configured security posture. Their use of high-quality web proxies, globally redundant edge CDNs (Fastly), and modern encryption protocols makes their primary web systems highly secure. The only minor issues discovered include easily harvestable employee email patterns and public access to old technical documents. Fixing these minor exposures will further shrink their public attack surface.

14. References

- ICANN Global Domain Registry WHOIS Information System
- Shodan Search Engine for Internet-Connected Devices (`hostname:mozilla.org`)
- VirusTotal Threat Intelligence Core Platform
- LinkedIn Business Directory & Company Insights Hub
- Wayback Machine Public Website History Archive





15. Internship Reflection

Completing this final capstone project has provided critical practical exposure to passive reconnaissance within enterprise environments, serving as an ideal conclusion to my OSINT internship at DevShieldX. This project demonstrated that open-source intelligence is far more than basic data gathering; it is a sophisticated capability required to map an organization's digital footprint and uncover hidden security risks.

Utilizing industry-standard threat intelligence platforms like Shodan, VirusTotal, Anymail Finder, and advanced Google Dorking against a target like Mozilla Corporation was an invaluable technical experience. It highlighted how fragmented digital artifacts—such as corporate email patterns, indexed legacy PDFs, and open network ports—can be systematically aggregated to map out a company's internal infrastructure.

This assignment has significantly enhanced my technical proficiency and analytical confidence in executing infrastructure intelligence workflows. Platforms like Shodan that initially appeared complex have now become tools I can comfortably use to extract intelligence and compile actionable security reports. This entire journey at DevShieldX has successfully bridged academic concepts with active security workflows, inspiring me to pursue a career in cybersecurity and threat intelligence.